

数据库原理(2020-2021-3)

标准答案与评分标准

1.数据冗余会造成存储浪费，造成数据不一致，带来版本控制问题。5‘

数据冗余的优点有：对数据库恢复，数据冗余是必须的。5‘

2.

	主键	外键
sailors	Sid 1‘	Master 2‘
boats	Bid 1‘	
reserves	Sid, day 3‘	Sid, bid 每个 1.5‘

3. (答案不唯一)

(1)	<pre>SELECT sailors.sname, reserves.day FROM sailors INNER JOIN reserves ON reserves.sid = sailors.sid WHERE reserves.bid = '106'; 或 SELECT sailors.sname, reserves.day FROM sailors, Reserves Where reserves.sid = sailors.sid and reserves.bid = '106'; 等价形式均可</pre>
(2-1)	<pre>SELECT sailors.sname FROM sailors, (SELECT reserves.sid FROM reserves, boats where reserves.bid=boats.bid and boats.color='red' GROUP BY sid HAVING count(*)>1) AS t1 where t1.sid = sailors.sid</pre>
(2-2)	<pre>SELECT sname FROM sailors WHERE sid in (select sid from reserves X, boats where reserves.bid=boats.bid and color='red' and exists(select * from reserves, boats where reserves.bid=boats.bid and color='red' and sid=X.sid and day!=X.day)); 注：写法不唯一，但两种写法都采用 Group by 子句，写的均正确，亦需扣 1-2 分。</pre>
3	<pre>预定过所有蓝色船只的水手 Select DISTINCT s.sid from reserves,boats Where reserves.bid=boats.bid and color='blue' Group by sid Having count(distinct bid)=(select count(*) from boats where color='blue');</pre>
4	<pre>SELECT sailors.rating, max(t.c)</pre>

	FROM (SELECT sid, count(*) AS c FROM reserves GROUP BY sid) AS t, sailors where t.sid=sailors.sid GROUP BY sailors.rating
--	--

4.

C 版本核心部分

Declare Cursor CR1

As

select reserves.bid, reserves.day, boats.bname from reserves, boats where reserves.bid=boats.bid
and sid=:id and bid=:bid

...

Open cursor CR1

While(true)

{

...

Fetch ... into...

...

}

Close CR1

需要有定义游标，打开游标，取数，关闭游标的语句，核心语句不全的，逐条扣 1-2 分！

Java 版核心部分

Report(int id){

 //建立数据库连接 ...

 String sql =.....

 Statement.executeSql(sql);

}

需要有传递水手编号、船编号参数的语句；实例化 **resultset** 类的语句，以及逐条处理数据的语句！

核心语句不全的，逐条扣 1-2 分！

5.

Create trigger checkReserves

Before insert on reserves (Before 3')

Referencing New as N

For each row

When (GetMon(N.say) in {7,8}) and (N.sid not in (select sid from reserves group by sid having
count(*)>9))

Rollback.

6.

(1) Checkpoint 用于数据库发生系统失效时，避免大量的无效 redo。

(2) 可以减少，具体与取 CP 的周期有关。

(3) 取后备副本后可以清空。

7.

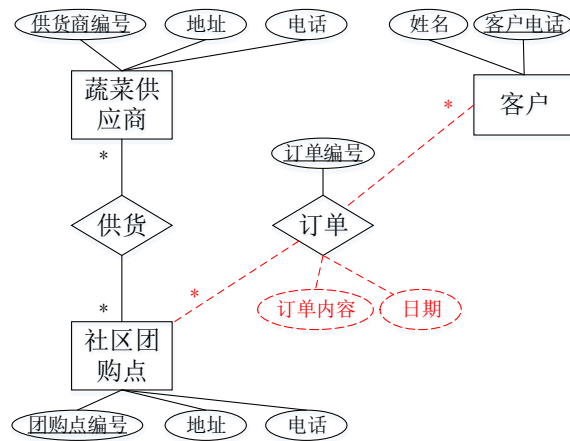
(1) 并发事务的正确性准则是可串行性。

(2) S 和 U 锁相容，可以提高并发度；U 和 U 锁相容，造成死锁。

(3) 从降低管理锁代价角度，(S,X) 锁更好

8.

(1)



(2)

